Part I: Given State of Model

The code for Part 1 was pulled directly from the course repository and is shown below in Figure 1. The initial model contains a simplified treatment of the electrochemistry in the anode where charge transfer reactions are only allowed at the electrolyte interface. This produces the flat potential distribution observed in the original results.

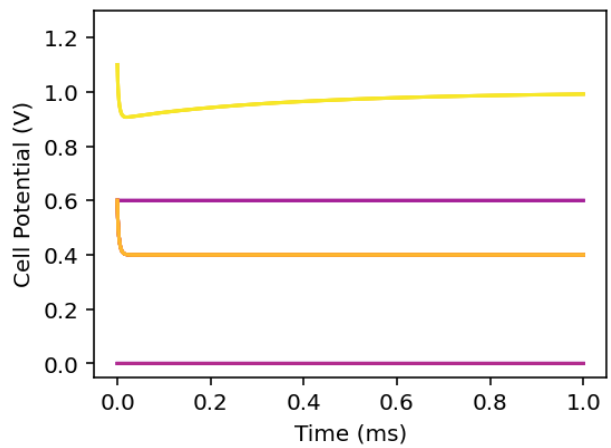


Figure 1: Initial model output from the provided solution.

Part II: Anode Metallic Electric Potential

From the original model, the code was modified to implement the following:

Charge neutrality was enforced across both the ionic and electronic conducting phases according to:

$$\nabla \cdot i_{io} + \nabla \cdot i_{el} = 0$$

which was discretized as:

$$i_{io,i} - i_{io,e} + i_{el,i} - i_{el,e} = 0$$

The ionic and electronic currents were calculated using Ohm's law:

$$i = -\sigma \nabla \phi$$

and discretized as:

$$i_m = \sigma_m^{eff} \frac{\phi_{m,0} - \phi_{m,1}}{\Delta y}$$

where the effective conductivity was determined using the Bruggeman correction:

$$\sigma_m^{eff} = \epsilon_m^{(1+n_{Brugg})} \sigma_m$$

Part III: Anode electrolyte phase potential

Faradaic current generation was implemented in every anode control volume using Butler-Volmer kinetics:

$$i = i_0 \left[\exp\left(-\frac{\beta n F \eta}{RT} \right) - \exp\left(\frac{(1-\beta) n F \eta}{RT} \right) \right]$$

with overpotential defined as:

$$\eta = \phi_{an,el} - \phi_{an,io} - U$$

Double Layer Charging

The ionic phase potential evolution was calculated using the double layer capacitance relationship:

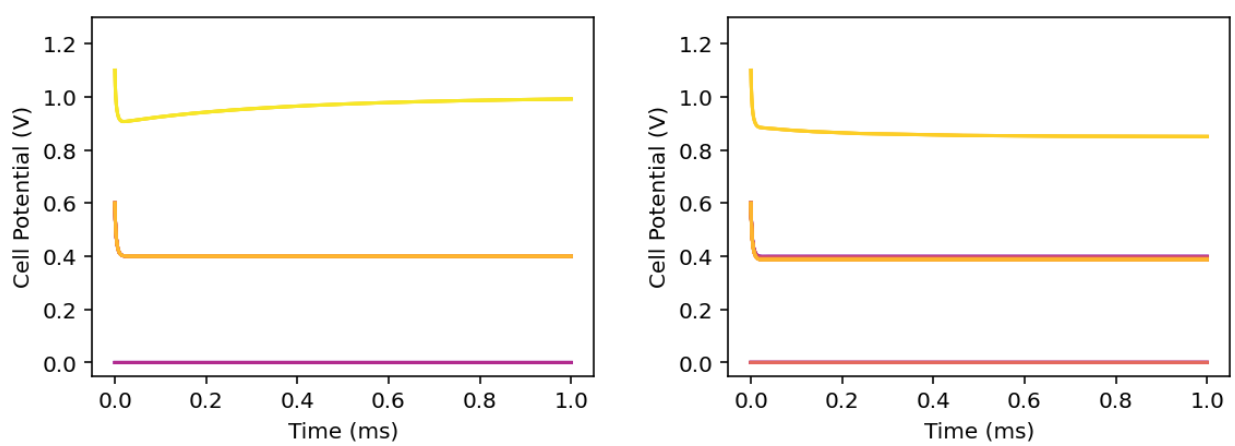
$$\frac{d\phi_{an,io}}{dt} = \frac{i_{dl}}{C_{dl}}$$

where the double layer current was determined from local charge conservation:

$$i_{dl} = i_{el,e} - i_{Far} - i_{el,i}$$

Part III: Verification of Updated Model

After implementing the distributed electrochemistry equations, the updated model produced the results shown in Figure 2.



(a) $i_{ext} = 0$ A/cm^2 (b) $i_{ext} = 0.5$ A/cm^2

Figure 2: Updated SOFC model results after implementing distributed electrochemistry equations.

A zoomed view of the ionic phase potential distribution in the anode is shown in Figure 3.

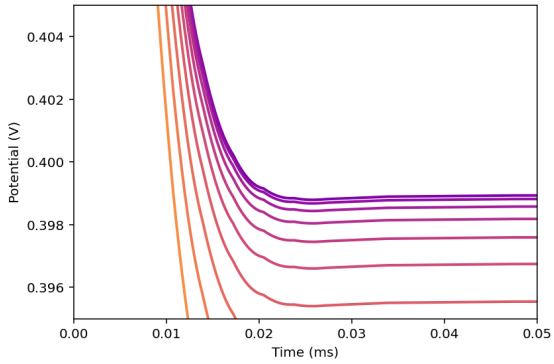
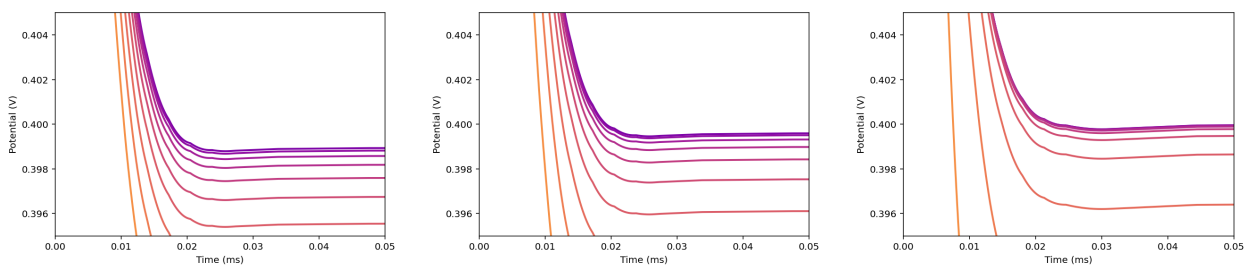


Figure 3: Zoomed ionic phase potential distribution in the anode.

The ionic phase potential varies primarily near the electrolyte interface and becomes nearly constant farther into the anode. This indicates that most of the electrochemical activity is localized near the electrolyte membrane. Once the ionic potential stops changing significantly, the local overpotential becomes very small and Faradaic current generation effectively ceases. This defines the electrochemically active thickness of the anode.

Effect of Ionic Conductivity

The ionic conductivity was varied from the baseline value of 10 $\mathrm{S\,m^{-1}}$ to lower values of 5 $\mathrm{S\,m^{-1}}$ and 1 $\mathrm{S\,m^{-1}}$ in order to investigate the effect of ionic transport limitations on the electrochemically active thickness of the anode.



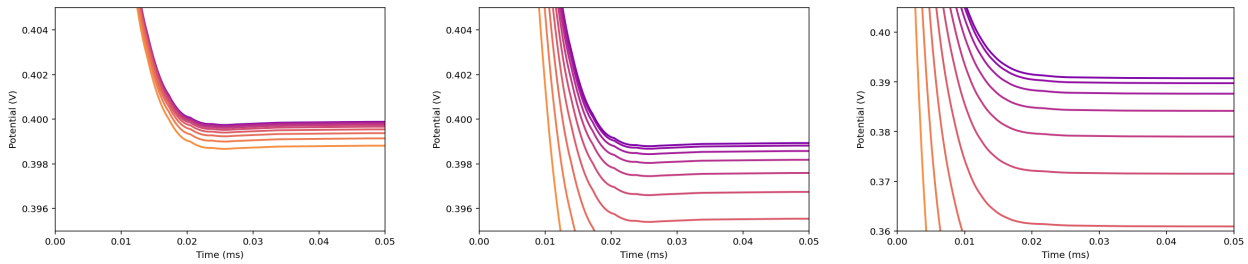
(a) $\sigma_{io} = 10$ S/m (b) $\sigma_{io} = 5$ S/m (c) $\sigma_{io} = 1$ S/m

Figure 4: Potential distributions for varying ionic conductivities.

As ionic conductivity decreases, the electrochemically active thickness becomes smaller and the ionic potential gradient becomes steeper near the electrolyte interface. Lower ionic conductivity increases ohmic losses within the ionic conducting phase, making it more difficult for ionic current to penetrate deeply into the anode. As a result, electrochemical reactions become increasingly localized near the electrolyte membrane. This behavior can be observed in Figure 4, where the potential curves separate more rapidly and flatten over a shorter distance as σ_{io} decreases, indicating a thinner active reaction region.

Effect of Exchange Current Density

The exchange current density was varied from the baseline value of 5×10^3 $\mathrm{A\,m^{-2}}$ to higher values of 5×10^4 $\mathrm{A\,m^{-2}}$ and 5×10^5 $\mathrm{A\,m^{-2}}$ in order to investigate the effect of reaction kinetics on the electrochemically active thickness of the anode.



(a) $i_{0,an} = 5 \times 10^3$ A/m^2 (b) $i_{0,an} = 5 \times 10^4$ A/m^2 (c) $i_{0,an} = 5 \times 10^5$ A/m^2

Figure 5: Potential distributions for varying anode exchange current densities.

Increasing the exchange current density increases the electrochemically active thickness of the anode. Higher exchange current density corresponds to faster reaction kinetics, allowing Faradaic reactions to occur with smaller overpotentials. This enables electrochemical activity to extend farther into the anode thickness and produces a broader active reaction region. This trend is visible in Figure 5, where increasing $i_{0,an}$ causes the potential gradients to extend farther across the anode thickness, corresponding to a larger electrochemically active region.